Chemistry Topic 4 — Acids, Bases, Salts and Carbohydrates

Sources baked in: NCERT Class 10 (Acids, Bases and Salts), Class 11 & 12 (Equilibrium; Biomolecules), Lucent General Science (Chemistry), Physics Wallah (PW) State PCS Foundation, Ghatna Chakra General Science (1990–2025 PYQs). **Weight:** ★★★ High — Recurring questions on baking soda vs washing soda formulas, Plaster of Paris vs gypsum water of crystallization, natural acid-source matches (oxalic in tomatoes, methanoic in ant stings), litmus color transitions, blood pH buffer range, and carbohydrate sweetness hierarchy. **Last verified:** September 2026

Current Affairs (2024–2026)

Year	Event	Core Concept & Must-Score Significance
2025–2026	**Global Ocean Acidification Monitoring**	Ocean absorption of anthropogenic CO_2 produces carbonic acid ($CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$), dropping surface ocean pH from pre-industrial 8.25 to ≈ 8.14. Even this 0.1 unit drop represents a $\approx 30\%$ increase in $[H^+]$ activity, severely impairing calcium carbonate ($CaCO_3$) calcification by coral reefs and molluscs.
2025	**UPPCS Prelims 2025**	Direct question on **pH Scale Inversion & Acid-Base Identification**: Keyed that statements claiming "Acids have pH > 7" or "Bases have pH < 7" are unequivocally false. Re-tested baking soda ($NaHCO_3$) decomposition in soda-acid fire extinguishers ($2NaHCO_3 + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O + 2CO_2 \uparrow$).
2024	**UPPCS Prelims 2024**	Question on **Litmus Paper Response**: Acid turns blue litmus paper red; base turns red litmus blue. Litmus is derived naturally from **Lichens** (symbiotic association of fungus and alga).
2024	**UPPCS / UKPCS Trends**	Formula: **Baking Soda is** $NaHCO_3$ (Sodium hydrogen carbonate); **Washing Soda is** $Na_2CO_3 \cdot 10H_2O$ (Sodium carbonate decahydrate); **Plaster of Paris is** $CaSO_4 \cdot \frac{1}{2}H_2O$; **Gypsum is** $CaSO_4 \cdot 2H_2O$.
2023	**Biomolecules & Sweeteners**	Testing of **Relative Sweetness Index**: **Fructose** is the sweetest naturally occurring sugar ($1.7\times$ sweeter than sucrose). Cane sugar (**Sucrose**) is a non-reducing disaccharide composed of α-D-glucose + β-D-fructose, hydrolyzing to yield optically inverted **Invert Sugar**.

Detailed Theory Notes

4.1 Acid-Base Theories & Mathematical Dynamics of the pH Scale

1. Theoretical Evolution

- **Arrhenius Concept (1884)**: Defined acids as substances producing H^+ ions in aqueous solution, and bases as producing OH^- ions. Limitation: Restricted strictly to aqueous systems and failed to

explain the basicity of ammonia (NH_3).

- **Brønsted-Lowry Concept (1923)**: Acid is a proton (H^+) donor; base is a proton (H^+) acceptor. Introduces **Conjugate Acid-Base Pairs** differing by exactly one proton (H^+):

$$\underset{\text{Acid}_1}{HCl} + \underset{\text{Base}_2}{H_2O} \rightleftharpoons \underset{\text{Acid}_2}{H_3O^+} + \underset{\text{Base}_1}{Cl^-}$$

- **Lewis Concept (1923)**: Broadest electronic definition:
- *Lewis Acid*: Chemical species capable of accepting an unshared electron pair (electrophile, e.g., $BF_3, AlCl_3, Fe^{3+}, H^+$).
- *Lewis Base*: Chemical species capable of donating an unshared electron pair (nucleophile, e.g., $: NH_3, H_2\ddot{O}:, OH^-, CN^-$).

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Arrhenius:      Acid = H⁺ generator in water     | Base = OH⁻ generator in water
Brønsted-Lowry: Acid = Proton (H⁺) donor         | Base = Proton (H⁺) acceptor
Lewis:          Acid = Electron pair acceptor    | Base = Electron pair donor
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2. Logarithmic pH Scale Dynamics

- Formulated mathematically by S.P.L. Sørensen (1909):

$$\text{pH} = -\log_{10}[H_3O^+]$$

- Because the scale is base-10 logarithmic, a decrease of 1 unit on the pH scale represents a **10-fold increase in hydrogen ion concentration**:

$$\frac{[H^+]_{\text{pH 2}}}{[H^+]_{\text{pH 5}}} = \frac{10^{-2}}{10^{-5}} = 10^3 = 1000 \times \text{ higher acidity}$$

- **Neutrality Definition**: Neutrality does NOT universally mean $\text{pH} = 7.0$; neutrality strictly means $[H^+] = [OH^-]$. At 25°C, $K_w = 1.0 \times 10^{-14}$, so $\text{pH} = 7.0$. At elevated physiological temperatures (37°C), $K_w \approx 2.4 \times 10^{-14}$, so neutral $\text{pH} \approx 6.8$.

4.2 Chemical Indicators & Titration Physics

1. Action of Acid-Base Indicators

Indicators are weak organic acids (HIn) or weak organic bases whose undissociated molecular form has a visibly distinct color from its ionized conjugate form (In^-):

$$HIn\ (\text{Color A}) \rightleftharpoons H^+(\text{aq}) + In^-\ (\text{Color B})$$

- In an acidic solution (high $[H^+]$), common-ion effect suppresses indicator dissociation, displaying Color A. - In an alkaline solution, added OH^- consumes H^+, shifting the equilibrium forward to display Color B.

2. Master Indicator Spectrum Table

Indicator Name	Nature / Source	Acidic Medium Color	Neutral Medium Color	Basic/Alkaline Medium Color	pH Transition Interval
Litmus	Natural (Lichen extract)	**Red / Pink**	Purple	**Blue**	$5.0–8.0$
Phenolphthalein	Synthetic (Phthalic anhydride + phenol)	**Colorless**	Colorless	**Magenta / Deep Pink**	$8.2–10.0$
Methyl Orange	Synthetic (Sodium sulfonate azo dye)	**Red / Orange-Red**	Orange	**Yellow**	$3.1–4.4$
Methyl Red	Synthetic azo dye	**Red**	Orange	**Yellow**	$4.4–6.2$
Turmeric (Haldi)	Natural rhizome (*Curcuma longa*)	**Yellow**	Yellow	**Reddish-Brown**	$7.4–9.2$
Red Cabbage	Natural anthocyanin pigment	**Red / Pink**	Purple	**Green to Yellow**	$2.0–12.0$

4.3 Household Salts & Industrial Chemistry

1. Baking Soda ($NaHCO_3$) vs. Washing Soda ($Na_2CO_3 \cdot 10H_2O$)

- **Solvay Industrial Process**:

$$NaCl + H_2O + NH_3 + CO_2 \rightarrow NaHCO_3 \downarrow + NH_4Cl$$

  Crude sodium hydrogen carbonate precipitates out because of its relatively low solubility in ammonium chloride brine.

- **Thermal Decomposition into Soda Ash**:

$$2NaHCO_3 \xrightarrow{\Delta} Na_2CO_3 + H_2O + CO_2 \uparrow$$

- **Hydration into Washing Soda**:

$$Na_2CO_3 + 10H_2O \rightarrow Na_2CO_3 \cdot 10H_2O \text{ (Washing Soda Decahydrate)}$$

- **Efflorescence Reaction**: In dry atmospheric conditions, washing soda loses 9 molecules of water spontaneously:

$$Na_2CO_3 \cdot 10H_2O \xrightarrow{\text{dry air}} Na_2CO_3 \cdot H_2O \text{ (Monohydrate)} + 9H_2O \uparrow$$

2. Plaster of Paris ($CaSO_4 \cdot \frac{1}{2}H_2O$) vs. Gypsum ($CaSO_4 \cdot 2H_2O$)

- **Thermal Regimes of Gypsum Calcination**: 1. *At* $100^\circ C$ *(*373 K*)*:

$$CaSO_4 \cdot 2H_2O \xrightarrow{373\text{ K}} CaSO_4 \cdot \frac{1}{2}H_2O \text{ (Plaster of Paris)} + 1\frac{1}{2}H_2O$$

  2. *Above* $200^\circ C$ *(*473 K*)*:

$$CaSO_4 \cdot 2H_2O \xrightarrow{>473\ \text{K}} CaSO_4\ (\text{Anhydrous "Dead Burnt Plaster"}) + 2H_2O$$

Dead burnt plaster absorbs water very slowly and has no hydraulic setting strength. 3. *At* $> 1000°C$:

$$2CaSO_4 \xrightarrow{>1000°C} 2CaO + 2SO_2 \uparrow + O_2 \uparrow$$

4.4 Carbohydrates, Sugars & Sweetener Dynamics

1. Structural Classification

- **Monosaccharides**:
- *D-Glucose*: Aldohexose; exists predominantly as a 6-membered pyranose ring (α-D-glucopyranose and β-D-glucopyranose).
- *D-Fructose*: Ketohexose; exists predominantly as a 5-membered furanose ring (α-D-fructofuranose and β-D-fructofuranose).
- **Disaccharides**:
- *Sucrose*: Composed of α-D-glucose $+$ β-D-fructose linked by an $\alpha - 1, \beta - 2$ glycosidic bond.
- *Lactose*: Composed of β-D-galactose $+$ β-D-glucose linked by a $\beta - 1, 4$ glycosidic bond.
- *Maltose*: Composed of two α-D-glucose molecules linked by an $\alpha - 1, 4$ glycosidic bond.

2. Inversion of Cane Sugar (Invert Sugar)

Sucrose is naturally dextrorotatory ($[\alpha]_D = +66.5°$). When cleaved by dilute acid or the enzyme invertase:

$$C_{12}H_{22}O_{11} + H_2O \xrightarrow{\text{invertase}} \underset{\text{Dextrorotatory } (+52.7°)}{\text{D-Glucose}} + \underset{\text{Strongly Levorotatory } (-92.4°)}{\text{D-Fructose}}$$

Because the levorotation of fructose exceeds the dextrorotation of glucose, the optical rotation of the resulting equimolar mixture is levorotatory ($-19.8°$). This inversion of the optical rotation gives **Invert Sugar** its name. Invert sugar is sweeter and more resistant to crystallization than sucrose, making it valued in confectionery and commercial honey.

Bilingual Terminology -- Acids, Bases, Salts and Sucrose

English Term	Hindi Term	Key Anchor
Acid (अम्ल)	अम्ल	Donates H+ (proton); pH < 7; sour taste; turns blue litmus red
Base (क्षार)	क्षार	Accepts H+ (proton); pH > 7; bitter taste, soapy feel; turns red litmus blue
Alkali (क्षारक)	क्षारक	Water-soluble base; NaOH, KOH, Ca(OH)2
pH (पीएच)	pH / पीएच	Negative log of H+ concentration; 0-14 scale; 7 = neutral
Neutralisation (उदासीनीकरण)	उदासीनीकरण	Acid + Base -> Salt + Water; exothermic

English Term	Hindi Term	Key Anchor
Salt (लवण)	लवण	Product of acid-base reaction; e.g., NaCl, CaCO3, CuSO4
Indicator (सूचक)	सूचक	Substance that changes colour with pH; litmus, phenolphthalein, methyl orange
Buffer (बफर)	बफर	Resists change in pH; e.g., blood pH 7.4 maintained by H2CO3/HCO3-
Bleaching Powder (विरंजक चूर्ण)	विरंजक चूर्ण / ब्लीचिंग पाउडर	Ca(OCl)Cl; used for bleaching, disinfecting water
Plaster of Paris (प्लास्टर ऑफ पेरिस)	पेरिस प्लास्टर	CaSO4.1/2H2O; sets hard by absorbing water; used in fractures, moulds
Washing Soda (धावन सोडा)	धावन सोडा	Na2CO3.10H2O; removes permanent hardness; used in glass, soap making
Baking Soda (बेकिंग सोडा)	मीठा सोडा / खाने का सोडा	NaHCO3; releases CO2 on heating; used in baking, antacids
Caustic Soda (दाहक सोडा)	दाहक सोडा / कास्टिक सोडा	NaOH; strong base; soap/paper making; drain cleaner
Vinegar (सिरका)	सिरका	Acetic acid solution (4-8% CH3COOH); sour taste
Hydrochloric Acid (हाइड्रोक्लोरिक अम्ल)	हाइड्रोक्लोरिक अम्ल	HCl; "muriatic acid"; in gastric juice; strong acid
Sulphuric Acid (सल्फ्यूरिक अम्ल)	गंधक का तेजाब	H2SO4; "king of chemicals"; strong acid; highly corrosive

One-Liner Revision -- Acids, Bases, Salts

#	Fast Fact
1	pH scale = 0-14; pH < 7 = acid; pH 7 = neutral; pH > 7 = base/alkaline
2	Strongest acid = HClO4 (perchloric); in chemistry; practically H2SO4 is "king"
3	Common acid in stomach (gastric juice) = **HCl** (hydrochloric acid)
4	Formic acid (HCOOH) = in ant sting; methanoic acid; simplest organic acid
5	Acetic acid (CH3COOH) = vinegar; ethanoic acid
6	Citric acid = lemons, oranges; ascorbic acid = Vitamin C
7	Lactic acid = sour milk (curd); produced by Lactobacillus
8	Oxalic acid = in spinach/tomato leaves; kidney stone-causing
9	Sulphuric acid (H2SO4) = "King of Chemicals"; most produced industrial chemical

#	Fast Fact
10	Aqua Regia = 3 parts HCl + 1 part HNO3; dissolves gold and platinum
11	Litmus = turns red in acid, blue in base; from lichen (Rocella tinctoria)
12	Phenolphthalein = colourless in acid; pink/red in base
13	Methyl orange = red in acid; yellow in base
14	NaOH (Caustic soda) = used in soap-making, paper, textile
15	Ca(OH)2 (slaked lime / calcium hydroxide) = whitewash; water treatment; mortar
16	Baking soda (NaHCO3) = antacid for acidity; releases CO2 in baking
17	Washing soda (Na2CO3.10H2O) = removes permanent hardness; laundry
18	Plaster of Paris (CaSO4.1/2H2O) = made from gypsum (CaSO4.2H2O) by heating
19	Bleaching powder Ca(OCl)Cl = disinfects water; bleaches cloth/paper
20	Common salt (NaCl) = table salt; from sea water by evaporation or rock salt mines
21	Blood pH = 7.4 (slightly basic); maintained by bicarbonate buffer
22	Acid rain pH < 5.6 (H2SO4 + HNO3 from SO2 + NOx emissions)
23	Antacids = neutralise excess HCl in stomach; Mg(OH)2, Al(OH)3, NaHCO3
24	Hard water + soap = scum (Ca/Mg stearate); soft water lathers well
25	Soap is sodium salt of fatty acid; detergent = synthetic sulphonate/sulphate
26	Saponification = hydrolysis of fats/oils with NaOH/KOH to make soap + glycerol
27	Sugar (sucrose) = C12H22O11; invert sugar = glucose + fructose (after acid hydrolysis)
28	Glucose = C6H12O6; dextrose; simplest monosaccharide; brain's primary fuel
29	Fructose = fruit sugar; sweetest naturally occurring sugar
30	Lactose = milk sugar; disaccharide (glucose + galactose); lactose intolerance = deficiency of lactase